Assignment for Database Systems Course

- Team consists of 5-6 members
- Deadline for group and topic registration **: 11:59 PM, September 30, 2026 for all class**

Requirements for Assignment 1

1. Analyze and Describe the Data Requirements for the Selected Topic (3 points)

1. **Research related applications/systems (reference):**
   Provide the name of the applications/systems, the link, a business process analysis, and supporting survey evidence with images (0.5 points).
2. **Describe the proposed system:**
   - Describe the proposed application/system (based on the reference system), including the types of users and the main system functions (0.5 points).
   - Describe the entity types, attributes, and relationships (similar to the COMPANY database description in the lecture slides) (1.5 points).
   - Must not present in tables or bullet points.
3. **Describe semantic constraints** (that cannot be represented in (E-)ERD) for entity types, attributes, and relationships (0.5 points).

Note: The database must include user account management.

2. EERD Design (4 points)

- The (E-)ERD diagram must be clear. **Hand-drawn diagrams or external links will not be accepted.**

- The design must be consistent with the reference application/website and the description in Part 1.

3. Mapping to Database Schema (3 points)

General Requirements

- **Minimum number of semantic constraints (*) = number of students in the group × 2.**

- **Minimum number of tables after mapping ()** = number of students in the group × 3**
  (At least 12 tables if the group has fewer than 5 members).

  - Tables generated from multivalued attributes count only once.

  - Tables generated from each 1:1 or 1:N relationship: students must explain why creating a new table is appropriate for it to be counted.

  - Tables generated from superclass/subclass: only count entities that have their own attributes **and** relationships.

- The E-ERD must include all entity types, relationships, and attributes covered in class:

- Entities, weak entities, superclass, subclass (union type is optional).
- Attributes: single-valued, multivalued, derived, composite (complex attributes are optional).
- Relationships: 1:1, 1:N, N:M, identifying relationships, recursive relationships (higher-order relationships, i.e., ternary or above, are optional).

Report Schedule and Format

- **Report for Assignment 1 (BTL1):**
  - Submit the report as a PDF file on the LMS of your lab class (the instructor will open the submission portal).
  - **Deadline:** 11:59 PM, Wednesday, October 14, 2026 (for CC04), 11:59 PM, Friday, Thursday October 15, 2026 for A01.
  - Submit a printed copy of the report and present the contents in class, including the (E)-ERD, to the instructor (the schedule depends on each instructor).

Suggested Topics

1. **E-commerce**
   - Marketplaces: Tiki, Lazada, Shopee
   - B2C: an online store or retail chain such as Nguyenkim, Fahasa, Thienlong, Hoayeuthuong, online pharmacy apps like Long Châu, Pharmacity, etc. (Choose a field such as fashion, footwear, food, books, etc.)
2. **E-learning**
   - Learning Management Systems (e.g., LMS at HCMUT)
   - Online learning websites: Udemy, Coursera, etc.
3. **Logistics**
   - Delivery of food, goods, freight, etc. (e.g., Ahamove, Grab Express)
   - Functions: placing delivery orders, choosing transportation methods, payment, shipment management, order tracking, customer care (loyalty points, promotions), administrative functions (vehicles, drivers, shipments, customers, etc.)
4. **Ticket Booking Systems**
   - Movie theaters, bus companies, events: CGV, Vexere, Futabus, Ticketbox, etc.
5. **Task/Expense/Asset Management for Individuals or Teams**
   - Examples: Jira, Trello, Money Lover, Misa, etc.
6. **Restaurant and Café Management**

- Examples: Ipos, KiotViet, etc.

7. **Job and Recruitment Platforms**

    - Examples: Itviec, Vietnamworks, etc.

Guidelines for Implementation

Step 1. Choose a field/topic of interest.
Step 2. Find existing related systems to explore (preferably those that allow easy account registration and free access to features). Select 1–2 systems that provide the most information and features as references.
Step 3. Study their features: analyze displayed data, input forms, reports, etc.
Step 4. Research additional related systems and references to infer more functions and data requirements.
Step 5. Specify the proposed system: describe its functions and complete data storage requirements.
Step 6. Complete the requirements of Major Assignment 1.

Notes on System Functions and Data Requirements

- The analysis must involve at least **two actors** (e.g., end-user and administrator).

Examples:

- **E-commerce (B2C):** buyer and store administrator.
- **E-commerce (C2C):** buyer and seller.
- Functions may include product information display, shopping cart, payment, shipping (by system shippers or third parties), order management, customer care (loyalty points, promotions, gifts), product/store reviews and comments, store information display, and administrative functions.
- **Online recruitment platforms:** candidate and employer.
    - For candidates: registration, profile creation, application management, etc.
    - For employers: registration, recruitment fee payment, job posting management, applicant review, result notifications, etc.
    - Job postings: open, closed, employer details, application submission, result checking.
- **E-learning:** learner and instructor.
- (and so on for specific features).